

P.Murali Krishna
(192525178)

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Scenario-Based Task (High)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.	BL3 – Apply	5
2	A hospital wants to track patient appointments. Design the relational schema and write SQL to find doctors with more than 10 appointments this month.	BL3 – Apply	5
3	In an e-commerce scenario, write SQL queries using sub-queries to find products that have never been ordered.	BL3 – Apply	5

4	For a university database, write relational algebra expressions to find students enrolled in both 'DBMS' and 'OS' courses.	BL3 – Apply	5
5	Given a payroll scenario, create a view showing employees whose salary is below department average and write a query to update their salary by 10%.	BL6 – Create	5
6	In a banking scenario, apply JOIN operations to retrieve customer account details, transaction history, and branch information.	BL3 – Apply	5
7	A retail store needs daily sales summary. Write SQL with GROUP BY and HAVING to report total sales per category exceeding Rs. 10,000.	BL3 – Apply	5
8	For a library system scenario, construct tuple relational calculus expressions to find members who borrowed books from all genres.	BL3 – Apply	5
9	Given an inventory management scenario, define CHECK and FOREIGN KEY constraints and demonstrate their enforcement through SQL.	BL3 – Apply	5
10	Design a relational schema for an HR system and create materialized views for monthly attendance and performance summaries.	BL6 – Create	5

Code of Conduct:

*I P.Murali Krishna (192525178) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Scenario based assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q1. School Management System – Students with Class Teacher & Grade (5 Marks)

Relations

```

Student(SID PK, Name, ClassID FK, DOB)
Class(ClassID PK, ClassName, TeacherID FK)
Teacher(TeacherID PK, Name, Subject)
Grade(SID FK, Subject, Marks, GradeLetter)
SELECT S.Name AS StudentName, T.Name AS ClassTeacher,
       G.Subject, G.GradeLetter
FROM Student S
JOIN Class C ON S.ClassID = C.ClassID
JOIN Teacher T ON C.TeacherID = T.TeacherID
JOIN Grade G ON S.SID = G.SID
ORDER BY S.Name, G.Subject;
```

Q2. Hospital – Doctors with more than 10 appointments this month (5 Marks)

Schema

```

Doctor(DoctorID PK, Name, Specialization, Phone)
```

```

Patient(PatientID PK, Name, DOB, Gender, Phone)
Appointment(ApptID PK, DoctorID FK, PatientID FK, ApptDate, Status)
SELECT D.DoctorID, D.Name, COUNT(A.ApptID) AS ApptCount
FROM Doctor D
JOIN Appointment A ON D.DoctorID = A.DoctorID
WHERE MONTH(A.ApptDate) = MONTH(CURRENT_DATE)
  AND YEAR(A.ApptDate) = YEAR(CURRENT_DATE)
GROUP BY D.DoctorID, D.Name
HAVING COUNT(A.ApptID) > 10;

```

Q3. E-commerce – Products that have never been ordered (5 Marks)

```

-- Method 1: NOT IN
SELECT ProductID, ProductName, Price
FROM Product
WHERE ProductID NOT IN (SELECT DISTINCT ProductID FROM OrderItem);

-- Method 2: NOT EXISTS (safer with NULLs)
SELECT P.ProductID, P.ProductName, P.Price
FROM Product P
WHERE NOT EXISTS (
  SELECT 1 FROM OrderItem OI WHERE OI.ProductID = P.ProductID
);

-- Method 3: LEFT JOIN
SELECT P.ProductID, P.ProductName
FROM Product P
LEFT JOIN OrderItem OI ON P.ProductID = OI.ProductID
WHERE OI.ProductID IS NULL;

```

Q4. University – Students enrolled in both DBMS and OS (Relational Algebra) (5 Marks)

```

 $\pi$  Name ( $\sigma$  CourseName = 'DBMS' (Student  $\bowtie$  Enrolled  $\bowtie$  Course))
 $\cap$ 
 $\pi$  Name ( $\sigma$  CourseName = 'OS' (Student  $\bowtie$  Enrolled  $\bowtie$  Course))

```

Alternatively using SID intersection then join back to Student for names.

Q5. Payroll – View of employees below dept average + 10% raise (5 Marks)

```

CREATE VIEW BelowAvgSalary AS
SELECT E.EmpID, E.Name, E.Salary, E.DeptID, D.DeptName
FROM Employee E
JOIN Department D ON E.DeptID = D.DeptID
WHERE E.Salary < (
  SELECT AVG(E2.Salary) FROM Employee E2

```

```
WHERE E2.DeptID = E.DeptID
);
```

```
UPDATE Employee
SET Salary = Salary * 1.10
WHERE EmpID IN (SELECT EmpID FROM BelowAvgSalary);
```

Note: Some DBMS restrict updates through complex views; update the base table using the same condition if needed.

Q6. Banking – JOIN for customer, account, transaction, branch (5 Marks)

```
SELECT C.CustomerID, C.Name, A.AccountNo, A.Balance,
       T.TransID, T.TransType, T.Amount, T.TransDate,
       B.BranchName, B.City
FROM Customer C
JOIN Account A ON C.CustomerID = A.CustomerID
JOIN Transaction T ON A.AccountNo = T.AccountNo
JOIN Branch B ON A.BranchID = B.BranchID
ORDER BY C.Name, T.TransDate DESC;
```

Q7. Retail – Daily sales summary with GROUP BY / HAVING (5 Marks)

```
SELECT Category, SUM(Amount) AS TotalSales,
       COUNT(*) AS NumTransactions
FROM Sales
WHERE SaleDate = CURRENT_DATE
GROUP BY Category
HAVING SUM(Amount) > 10000
ORDER BY TotalSales DESC;
```

Q8. Library – Members who borrowed books from ALL genres (Tuple RC) (5 Marks)

Assume: Member(MID, Name), Book(BID, Title, Genre), Borrow(MID, BID)

```
{ m.Name | Member(m) ∧
  ∀ g ( ∃ b (Book(b) ∧ b.Genre = g) →
    ∃ br (Borrow(br) ∧ br.MID = m.MID ∧
      ∃ b2 (Book(b2) ∧ b2.BID = br.BID ∧ b2.Genre = g)) ) }
```

Meaning: For every genre that exists, the member has borrowed at least one book of that genre.

Q9. Inventory – CHECK & FOREIGN KEY Constraints (5 Marks)

```

CREATE TABLE Category (
  CategoryID INT PRIMARY KEY,
  CategoryName VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE Product (
  ProductID INT PRIMARY KEY,
  ProductName VARCHAR(100) NOT NULL,
  CategoryID INT,
  Quantity INT CHECK (Quantity >= 0),
  UnitPrice DECIMAL(10,2) CHECK (UnitPrice > 0),
  ReorderLevel INT CHECK (ReorderLevel >= 0),
  CONSTRAINT fk_category FOREIGN KEY (CategoryID)
    REFERENCES Category(CategoryID)
    ON DELETE RESTRICT
    ON UPDATE CASCADE
);

-- Fails CHECK: INSERT INTO Product VALUES (1,'Pen',1,-5,10.00,20);
-- Fails FK: INSERT INTO Product VALUES (2,'Pencil',999,100,5.00,50);

INSERT INTO Category VALUES (1, 'Stationery');
INSERT INTO Product VALUES (1, 'Pen', 1, 200, 10.00, 50);

```

Q10. HR System Schema + Materialized Views (5 Marks)

Schema

```

Employee(EmpID PK, Name, DeptID FK, Designation, JoinDate)
Department(DeptID PK, DeptName)
Attendance(AttID PK, EmpID FK, AttDate, Status, HoursWorked)
Performance(PerfID PK, EmpID FK, ReviewPeriod, Rating, Comments)

```

Materialized Views

```

CREATE MATERIALIZED VIEW MV_MonthlyAttendance AS
SELECT E.EmpID, E.Name, D.DeptName,
       TO_CHAR(A.AttDate, 'YYYY-MM') AS Month,
       COUNT(CASE WHEN A.Status = 'Present' THEN 1 END) AS PresentDays,
       COUNT(CASE WHEN A.Status = 'Absent' THEN 1 END) AS AbsentDays,
       SUM(A.HoursWorked) AS TotalHours
FROM Employee E
JOIN Department D ON E.DeptID = D.DeptID
JOIN Attendance A ON E.EmpID = A.EmpID
GROUP BY E.EmpID, E.Name, D.DeptName, TO_CHAR(A.AttDate, 'YYYY-MM');

CREATE MATERIALIZED VIEW MV_MonthlyPerformance AS
SELECT E.EmpID, E.Name, D.DeptName, P.ReviewPeriod,
       AVG(P.Rating) AS AvgRating, MAX(P.Rating) AS MaxRating
FROM Employee E

```

```
JOIN Department D ON E.DeptID = D.DeptID
JOIN Performance P ON E.EmpID = P.EmpID
GROUP BY E.EmpID, E.Name, D.DeptName, P.ReviewPeriod;

REFRESH MATERIALIZED VIEW MV_MonthlyAttendance;
REFRESH MATERIALIZED VIEW MV_MonthlyPerformance;
```

I P.Murali Krishna (192525178) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Signature of the student: _____

A rectangular box containing a handwritten signature in blue ink. The signature appears to read "P. Murali Krishna".
